Labour Demand 2: Non-wage Benefits and Quasi-Fixed Costs

EC306- Wages and Employment

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Goals of This Section

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- Distinguish between workers and hours as an input into production
- Incorporate non-wage benefits into the labour demand model
- Also add costs that are independent of hours worked (quasi-fixed costs)
- Discuss the implications of quasi-fixed costs for labour demand

Background

Background

- So far we have discussed labour as a purely variable input paid a wage per hour worked
- In reality it is not so simple
    - Workers are sometimes paid other benefits besides wages
        - CPP, EI, , retirement benefits, paid vacation, sick leave, etc.
    - There are costs beyond the hourly wage
        - Hiring and training costs
        - Costs of dismissal
        - Non-wage benefits described above
- These benefits and costs complicate the labour demand decision

Non-Wage Benefits

Non-wage Benefits

- As discussed above, compensation typically includes more than just wages
- Common non-wage benefits include
  - Health insurance (beyond public health care)
  - Retirement benefits
  - Paid vacation and sick leave
  - Other perks (e.g. gym memberships, childcare, etc.)
- These benefits have value to workers
- They increased as a share of total compensation through the 90s and early 2000s

Non-wage Benefits

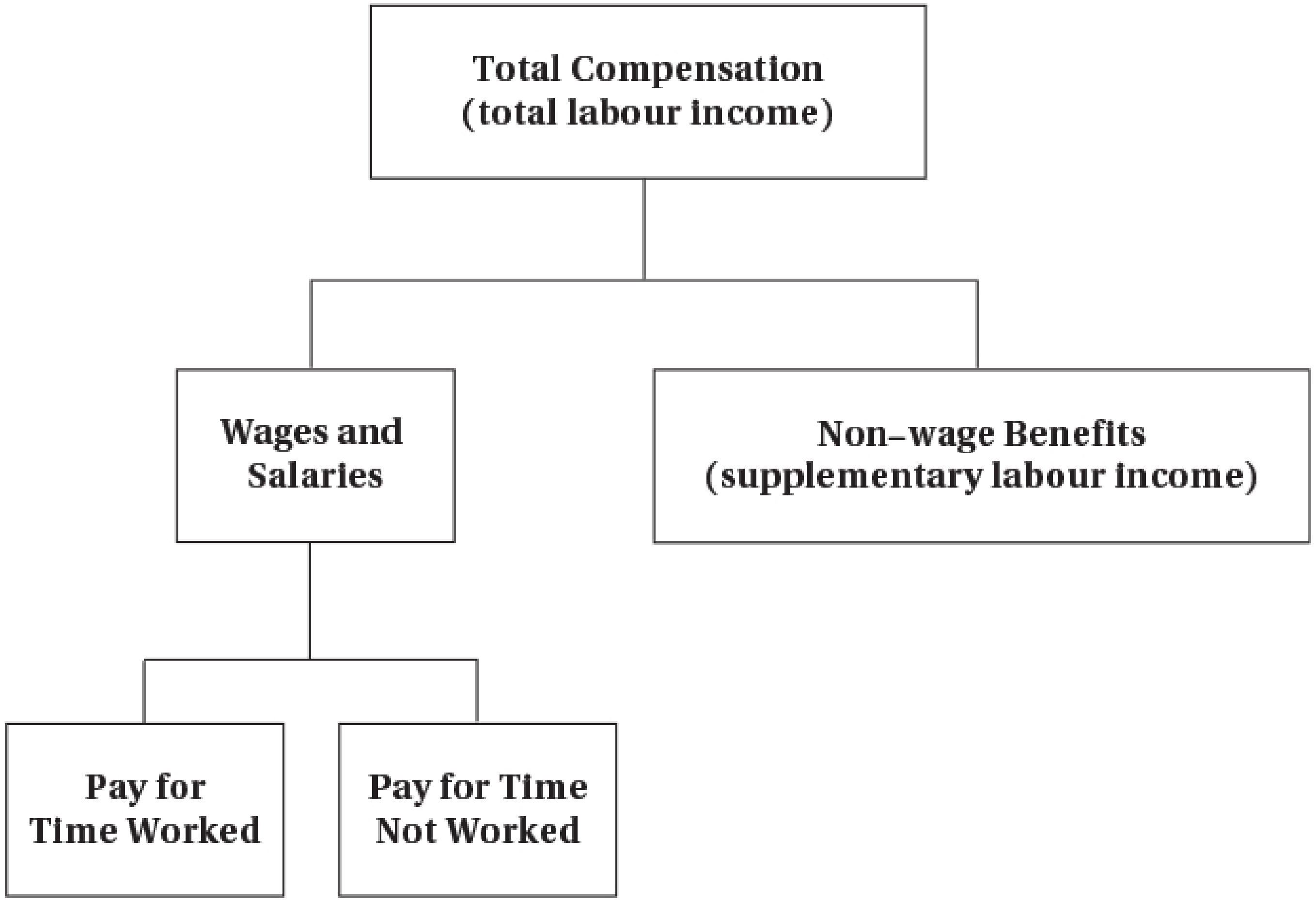


FIGURE 6.1 Components of Total Compensation

Total compensation includes wages plus non-wage benefits. The wage and salary component can be further divided into the part directly corresponding to time worked and that which does not depend on time worked (such as vacation pay).

Non-wage Benefits

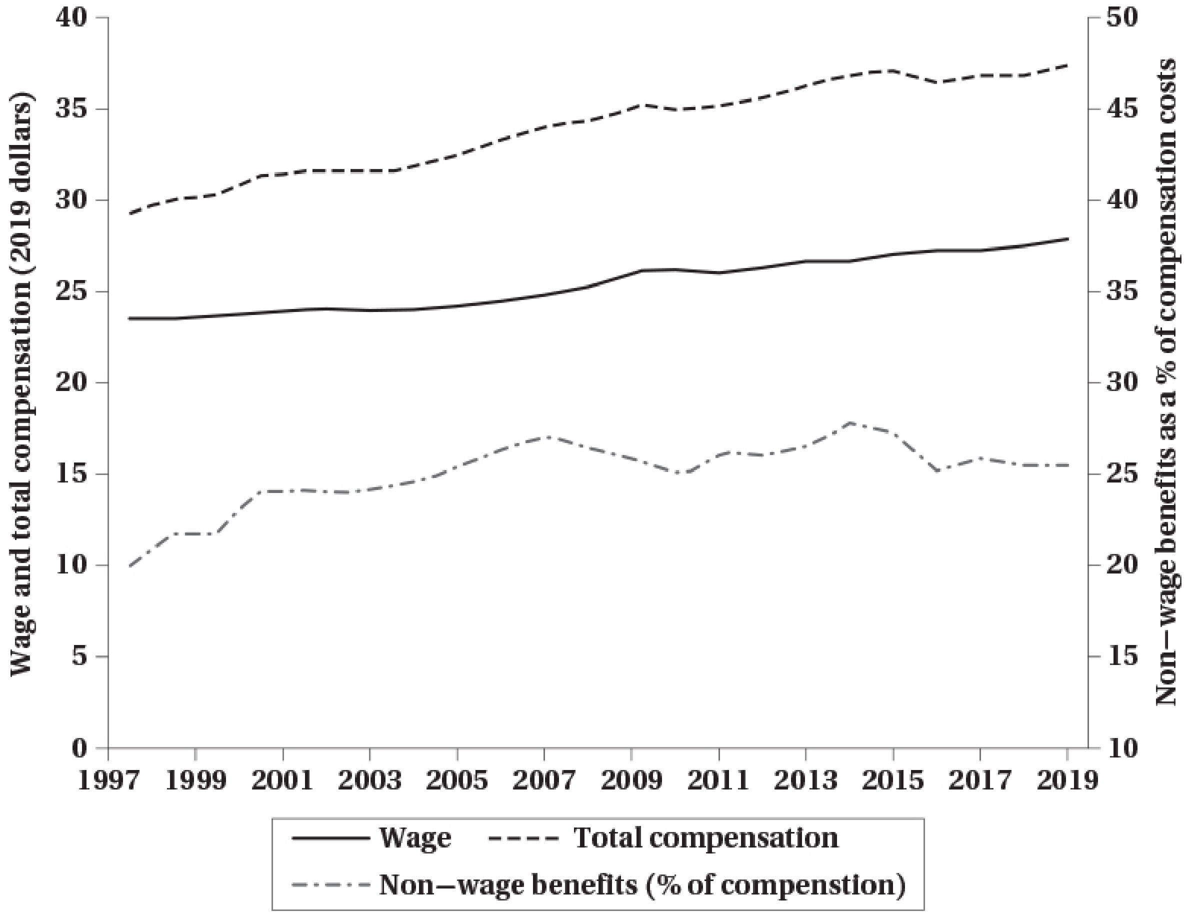


FIGURE 6.2 Total Compensation, Wage, and Non-wage Benefits

This figure shows the evolution of total compensation per hour, wages per hour, and non-wage benefits as fraction of total compensation. Total compensation and wages are expressed in dollars of 2019. Total compensation if growing faster than wages, reflecting the growing contribution of non-wage benefits to total compensation.

SOURCE: Statistics Canada table 36-10-0489-01 (total compensation from the System of National Accounts) and table 14-10-0064-01 (hourly wage from the Labour Force Survey).

Non-wage Benefits

- Why do firms pay these benefits instead of wages?
  - Sometimes they are not taxed
  - Economies of scale in providing benefits
    - Things like pensions and insurance are cheaper when provided to a group
    - Because it avoids adverse selection problems
  - Worker preferences
    - Some workers may prefer benefits to wages
    - E.g. risk-averse workers may prefer health insurance
  - Legal requirements

Non-wage Benefits

- Continued…
  - Better firm planning
    - Some benefits can reduce absenteeism
    - Contingency plans for worker illness or accidents through sick leave
  - Better for productivity
    - Subsidized meals keep people working
  - Deferred compensation
    - Retirement benefits can help retain workers
    - Workers may accept lower wages in exchange for future benefits
    - Acts as a screen for good workers

Non-wage Benefits

- Governments sometimes prefer (and mandate) them also
  - Firm pensions reduce need for public pensions
  - Better health outcomes reduce public health care costs
  - Improve the social safety net through sick leave and unemployment insurance
- With all these reasons, it is not surprising that non-wage benefits are common

Quasi-fixed Costs

Introduction

- We have been agnostic about whether labour is measured in workers or hours worked
- But there is an important distinction
  - Some costs are incurred per worker regardless of hours worked
  - Other costs are incurred per hour worked
- Costs that occur per worker but are relatively independent of hours worked are **quasi-fixed costs**
  - Pure quasi-fixed costs do not vary with hours worked at all
  - In practice, many costs are partially quasi-fixed
  - Some of these costs may be recurring (e.g. monthly EI premiums) and others one-time (e.g. hiring and training costs)

Examples of Quasi-fixed Costs

- Hiring costs
  - Recruiting, interviewing, and training are costly
  - Incurred when a worker is hired regardless of hours worked
  - Firms generally try to minimize hiring costs by retaining workers
- Termination costs
  - Severance pay, legal fees, and administrative costs
  - Initially depend on time with company, but have a maximum so they become fixed
  - Firms may be reluctant to lay off workers because of these costs

Examples of Quasi-fixed Costs

- Employer contributions to payroll taxes
  - EI and CPP are initially related to work hours
  - But they have a ceiling so they become quasi-fixed for high hours worked
- Other non-wage benefits
  - Employer contributions to life insurance, medical, dental plans
  - Often have a fixed component regardless of hours worked
- Note that if a cost always varies with hours worked it is not quasi-fixed
  - E.g. hourly wage, overtime premiums, per-hour health and safety costs, etc.

Effects of Quasi-fixed Costs

- Previously we saw that labour demand is determined by equating MRP_L to the wage rate
- The decision is similar with quasi-fixed costs, but now the cost of hiring a worker is higher
- There is also a dynamic component where firms amortize the costs over time
  - Firms may pay high training costs now
  - But expect the worker to remain with the firm to recoup those costs

Effects of Quasi-fixed Costs

- A profit maximizing firm will hire workers up to the point where

$$H + T + \sum_{t=1}^{N} \frac{w_t}{(1+r)^t} = \sum_{t=1}^{N} \frac{VMP_t}{(1+r)^t}$$

- Where
  - H = hiring cost
  - T = training cost
  - w_t = wage in period t
  - VMP_t = value of marginal product in period t
  - r = discount rate
  - N = expected tenure of worker

Effects of Quasi-fixed Costs

- This equation has several implications
  - Firms will be reluctant to hire workers for short periods if hiring costs are high
  - The wage does not have to equal the marginal product of labour in each period
  - Future productivity matters, but is discounted
  - Firms may be willing to pay more than the current marginal product if they expect future productivity to be high
- Related to the last point, when hiring costs are a factor future productivity must exceed the total wage

$$\sum_{t=1}^{N} \frac{VMP_t}{(1+r)^t} > \sum_{t=1}^{N} \frac{w_t}{(1+r)^t}$$

Effects of Quasi-fixed Costs

- This is similar to any other investment decision
  - Firms invest in workers when the present value of future returns exceeds the present value of costs
  - Hiring and training costs are like "upfront" costs of investment
  - Wages are like "ongoing" costs of investment
  - Future productivity is like the "returns" on that investment
  - You only make the investment if you expect to earn at least as much as you spend
- The next slide presents an example

Effects of Quasi-fixed Costs

	Costs					
Employment Duration (N)	Fixed (H + T)	Variable ($\sum W_t$)	Total	Benefits ($\sum VMP_t$)	Total Profit	Yearly Profit
1 year	\$15,000	\$25,000	\$40,000	\$30,000	−\$10,000	−\$10,000
2 years	15,000	50,000	65,000	60,000	−5,000	−2,500
3 years	15,000	75,000	90,000	90,000	0	0
4 years	15,000	100,000	115,000	120,000	5,000	1,250
5 years	15,000	125,000	140,000	150,000	10,000	2,000

Things Explained by Quasi-Fixed Costs

Overtime

- Firms sometimes work employees longer despite higher overtime wages
- Can be explained by high quasi-fixed costs of hiring additional workers
  - May be cheaper paying these higher wages to incurring hiring and training costs
  - Especially the case for high skill workers who require a lot of training

Temp and Gig Workers

- Firms do not always hire part- or full-time employees directly
- They sometimes use temp agencies, or “gig” workers
  - Laurier hires a cleaning firm and does not employ many janitors directly
  - Uber classifies drivers as independent contractors
- By doing this firms do not pay quasi-fixed costs

Layoffs

- During downturns in the economy firms sometimes keep workers around
- This can be explained by quasi-fixed costs of hiring and termination
  - Paying severance and incurring legal fees may be costly
  - Firms may prefer to keep workers on payroll if they expect a quick recovery
  - Firms may also be reluctant to lay off skilled workers due to high training costs
- This can lead to **labour hoarding** during downturns
- Also explains why highly educated workers tend to suffer less unemployment during recessions

Layoffs

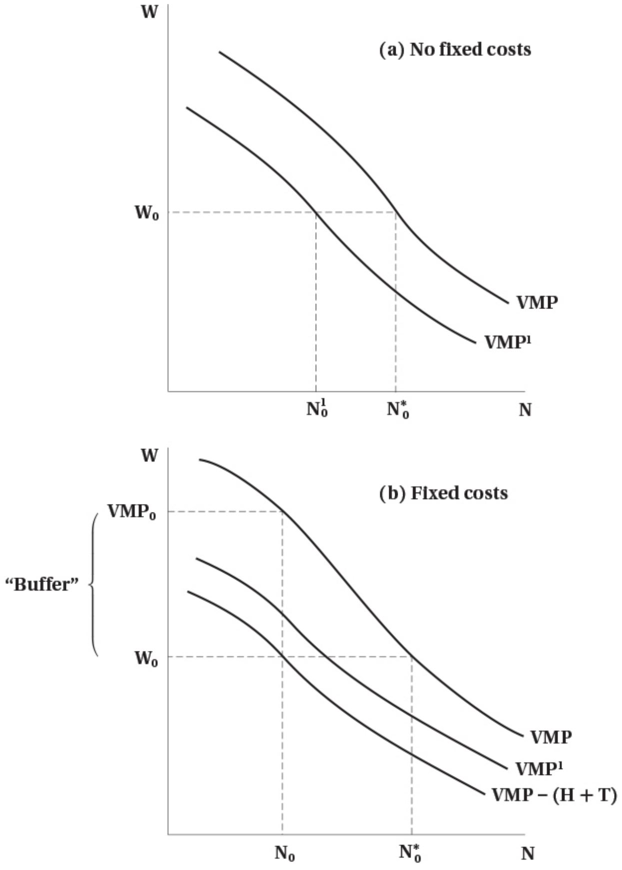
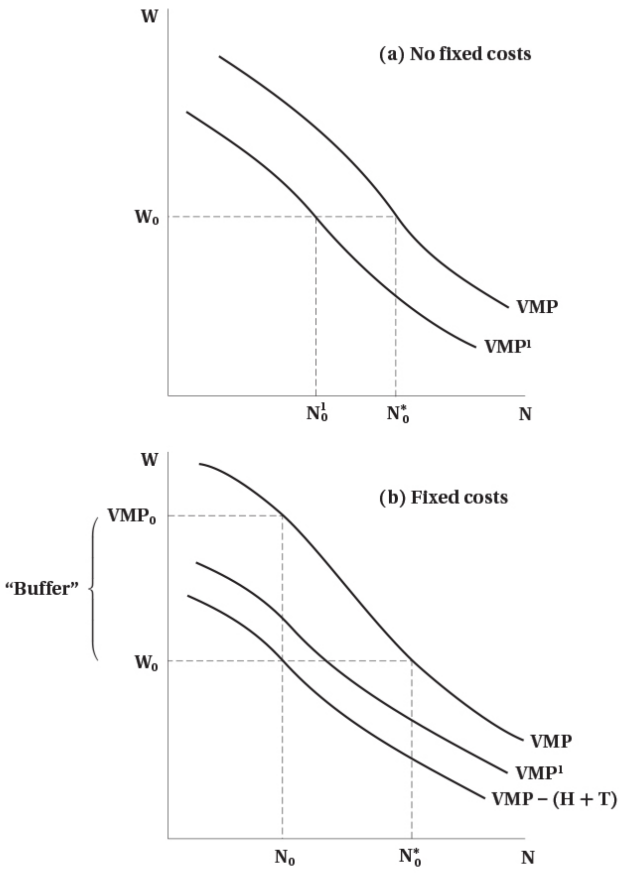


FIGURE 6.3 Nonrecurring Fixed Employment Costs and Changes in Labour Demand

In panel (a), a decline in market conditions leads to an inward shift of the VMP schedule, from VMP to VMP[1]. The profit-maximizing level of employment falls from N_0^* to N_0^1 at the prevailing wage of W_0. By contrast, if there are fixed costs of hiring workers, employment may not fall as much. First, the original level of employment will be at N_0, not N_0^*, as employers choose employment so that the wage equals the VMP net of hiring and training costs, VMP − (H + T). At N_0, the VMP (VMP_0) exceeds W_0 by a "buffer," H + T. As long as the new $VMP^1 > VMP - (H + T) = W_0$, it will not be optimal for the firm to change its employment level from N_0.

- Graph to left illustrates labour hoarding
- Top panel is without quasi-fixed costs
- Firm hires to point where $VMP = w$
- In economic downturn, VMP shifts down
  - Because of lower demand for output, and therefore price
- With the same wage, firms hire fewer workers

Layoffs



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- Bottom panel is with quasi-fixed costs
- Quasi-fixed costs shift VMP curve down prior to downturn
  - Firm hires to point where $VMP - (H + T) = w$
  - Fewer workers are hired initially because of these costs
- Creates wedge between VMP and wage
  - VMP is now higher than wage
- After downturn, VMP shifts down
  - But because of wedge, firm may not change employment

Segmentation

- Hiring costs mean trained and untrained employees are different
  - Firms may be reluctant to hire untrained workers because of high training costs
- When quasi-fixed costs are high, firms to prefer to retain or hire trained workers
- Leads to a segmentation into protected and unprotected workers in labour market
- Also leads firms to work employees more intensively than hire new workers
  - Example: big law firms that work associates long hours instead of hiring new associates

Work Sharing

- Some individuals would embrace work sharing
  - Two or more people share a job
  - Hiring more people to each work less hours for the same total hours
- Quasi-fixed costs explain why firms do not engage more in this activity
  - Hiring costs make it costly to bring on additional workers
  - Firms prefer to have fewer workers working more hours each
  - Even if total hours worked is the same

Work Sharing

Introduction

- In this section we expand on work sharing
- In Canada work sharing is a government program that subsidizes employee wages during downturns
  - Administered through Employment Insurance
  - Allows firms to reduce hours of employees instead of laying them off
  - Employees form a “work sharing group” and agree to reduce hours
  - Maximum duration of 26 weeks, with possible extensions
- The key idea is to avoid layoffs and employ more people
  - This is why the government is involved

Introduction

- There are other possible work sharing schemes
  - Delayed entry into the labour force
  - Early retirement
  - Short work weeks
  - Reduced overtime
  - Increased vacation time
- Do any of these schemes increase the number of jobs?

Overtime Restrictions

- As noted above, one way to share work is to restrict overtime
  - Firms do not like this because as noted it could increase costs
- Some governments have enacted legislation to prevent excessive overtime
- It is not clear that these laws increase employment
  - Firms may not hire more workers
  - The increased costs reduce overall demand for labour
  - New workers may not be as productive as existing workers, so firms may not want to hire them
  - There may not be enough employable workers available

Overtime Restrictions

- Also the possibility that part of reason for work sharing is based on a fallacy
  - **Lump of Labour Fallacy** is that there is a fixed number of jobs in the economy
  - Leads people to believe that employees working longer hours are taking jobs away from others
  - But in reality the number of jobs is not fixed
  - As people become employed, the economy grows, and more jobs are created
  - Fallacy is commonly used to argue against immigration
- Studies show that restricting work hours does not lead to increased employment
- But may still be useful in the short term in economic downturns

Subsidized Work Sharing

- As noted, Canada subsidizes work sharing programs for economic downturns
- Main goal is to keep people employed during recessions
- But would be good if it also protect jobs longer term
- Evidence suggests that
  - Many firms use the program
  - It does help keep people employed during downturns
  - But there is little evidence that it increases employment longer term
    - Employees were laid off when subsidy ran out

Employee Demand for Work Sharing

- We have been discussing work sharing from persepective of employer
- Employees need to also buy in for it to be successful
- It is not clear that they do in many cases
- Example: union jobs
  - Unions often resist work sharing programs
  - Prefer to keep full-time jobs with benefits
  - May see work sharing as a threat to job security
  - May also see it as a way for employers to reduce wages and benefits

Summary

Summary

- In this section we expanded the labour demand model
- We incorporated non-wage benefits and quasi-fixed costs
- Non-wage benefits are common and have many justifications
- Quasi-fixed costs complicate the labour demand decision
- These costs can explain many labour market phenomena
- Work sharing is one possible response to quasi-fixed costs
- But it is not clear that work sharing increases employment longer term